

ChatHealthy.ai

ARCH-001 Refactor Design

Consolidated Design Document

Iterations 1, 3, and 5 merged into final specification

Date: 2026-03-30

Claude Opus 4.6 (engineer) + GPT-5.3 (architect), 5 iterations, 2026-03-27 through 2026-03-30

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1. Executive Summary

Modular monolith refactor of FindCare backend from a 1,712-line main.py monolith to a facade-oriented domain architecture. Two business domains (FindCare, EvaluateCareQuality) with explicit service classes per UAT feature. HuggingFace reduced to endpoint + adapter only. Strangler Fig migration in 7 phases, zero downtime, deployable at every phase. Pydantic-enforced tool boundaries. Full Perplexity audit response covering all 18 findings.

2. Top-Level Architecture

Pattern: Modular Monolith (Facade-Oriented Domains)

Dependency Direction: Host -> Application -> Domain -> Infrastructure (no upward imports)

Layers

- Host Adapter (HuggingFace only -- app.py + HostChatAdapter)
- Application Layer (ChatOrchestrator + Facades + ToolRouter)
- Domain Layer (Business Services per UAT feature)
- Infrastructure Layer (MongoDB, External APIs, Embeddings, Config)

3. Business Components

3.1 FindCareFacade - Core provider discovery and specialty identification

Service Class	File	UAT Feature	Responsibility
ProviderSearchService	domain/find_care/provider_search_service.py	Provider Search (UAT Feature 1)	Vector search, query expansion, FIPS county mapping, re formatting
SpecialtyService	domain/find_care/specialty_service.py	Specialty Identification (UAT Feature 2)	

Facade Methods

- search_providers(criteria: ProviderSearchInput) -> ProviderSearchOutput
- identify_specialty(input: SpecialtyInput) -> SpecialtyOutput
- get_provider_location(npi: str) -> dict

3.2 EvaluateCareFacade - Clinical trials search and provider quality evaluation

Service Class	File	UAT Feature	Responsibility
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ClinicalTrialsService	domain/evaluate_care_quality/clinical_trials_service.py	Clinical Trials Search (UAT Feature 3)	ClinicalTrials.gov API integration, result filtering
ProviderDetailService	domain/evaluate_care_quality/provider_detail_service.py	Provider Detail (UAT Feature 8)	NPI lookup, external provider data aggregation

Facade Methods

- search_clinical_trials(input: ClinicalTrialsInput) -> ClinicalTrialsOutput
- get_provider_details(npi: str) -> ProviderDetailOutput

Cross-Domain Calls

- EvaluateCareFacade calls FindCareFacade.get_provider_location() for travel calculations

4. Shared Infrastructure Services

Service Class	File	UAT Feature	Responsibility
SafetyService	domain/shared/safety/safety_service.py	Safety Filter (UAT Feature 5)	IP locking, session safety state, pre-tool gating
EmergencyDetector	domain/shared/safety/emergency_detector.py	Emergency Detection (F-09 fix)	Config-driven keyword + embedding similarity detection
ConsentService	domain/shared/consent/consent_service.py	Consent Framework (UAT Feature 7)	Full two-tier HIPAA consent workflow -- see Section 4.1 Consent Framework Detail
LeadService	domain/shared/lead_capture/lead_service.py	Lead Capture (UAT Feature 6)	Contact record creation, consent integration
AboutService	domain/shared/content/about_service.py	About ChatHealthy (UAT Feature 4)	Static content loading for about intents
UnknownQuestionService	domain/shared/unknowns/unknown_question_service.py	Unanswerable Question Handling (UAT Feature 12)	De-identified logging of unanswerable questions
URLGuardian	domain/shared/url/url_guardian.py	URL Validation (UAT Feature 9)	Dual-mode URL validation (tool results + LLM text)

4.1 Consent Framework -- Full Workflow Design

HIPAA-aligned two-tier consent framework controlling what user data is stored and how. Consent is a full-time workflow -- not a flag. It governs the entire data lifecycle of a user session.

Trigger: Every 5th user message, system prompt instructs Claude to ask: 'Would you like someone from ChatHealthy.AI to follow up with you personally?'

Two-Tier Consent Flow

Tier 1: Verbatim Consent

Prompt: "May we save a verbatim transcript of this conversation with your contact details?"

If YES:

- consent_verbatim: True
- stored: Full chat_history (PII intact), email, name, notes, reason_for_contact
- evidence: Consent exchange itself is included in the stored transcript

If NO: Proceed to Tier 2

Tier 2: De-Identified Summary Consent

Prompt: "May we save a de-identified summary of this conversation instead?"

If YES:

- consent_verbatim: False
- consent_summary: True
- stored: LLM-generated summary passed through deIdentify() -- no PII, no chat_history
- process: Claude summarizes -> deIdentify() scrubs PII -> stored in notes field

If NO:

- consent_verbatim: False
- consent_summary: False
- stored: Contact fields only (email, name). No history, no summary.

Data Streams

Stream	Consent Required	Treatment	Collection
Questions (unanswerable)	False	Always de-identified before storage. No PII. No consent needed.	AboutUs.AboutSkip
Contact / Lead Capture	True	Two-tier consent governs what is stored. User chooses verbatim or summary.	AboutUs.lead

ConsentService Methods

- request_consent(session_id) -> ConsentState
- record_verbatim_consent(session_id, chat_history, contact_info) -> LeadRecord
- record_summary_consent(session_id, summary, contact_info) -> LeadRecord
- record_no_consent(session_id, contact_info) -> LeadRecord
- de_identify(text: str) -> str

Integration Points

- ChatOrchestrator: triggers consent prompt every 5th message
- LeadService: delegates to ConsentService for data handling
- System Prompt: RULE 7 defines the consent UX flow
- deIdentify(): shared utility used by ConsentService and UnknownQuestionService

Regulatory Notes

- Consent exchange itself must be preserved as evidence (included in verbatim transcript)
- De-identification must be applied BEFORE storage, not after
- Emergency records (BTC-003) must NEVER be deleted regardless of consent state
- Test data tagged with testdata=true for filtering

Lead Record Schema

- email: string
- name: string
- notes: string (summary if consent_summary=true, else empty)
- reason_for_contact: string
- consent_verbatim: boolean
- consent_summary: boolean or null
- chat_history: array (present ONLY if consent_verbatim=true)
- datetime: ISO string
- testdata: boolean

5. HuggingFace Surface Contract

Allowed Files

- app.py
- requirements.txt
- static/

Responsibilities

- HTTP routing (FastAPI app creation, CORS, endpoints)
- Request/response translation
- Stream relay
- NO business logic

Adapter Class: HostChatAdapter

Adapter File: adapter/host_chat_adapter.py

Entrypoint Final State

main.py: DELETED

Replacement: app.py at Code/ConversationalUX/FindCareChat/app.py

- FastAPI app initialization
- Route definitions (/chat, /health, /welcome)
- Delegates ALL logic to HostChatAdapter

6. Facade Design

FindCareFacade (application/facades/find_care_facade.py)

Methods:

- search_providers(criteria: ProviderSearchInput) -> ProviderSearchOutput
- identify_specialty(input: SpecialtyInput) -> SpecialtyOutput
- get_provider_location(npi: str) -> dict

Delegates to: ProviderSearchService, SpecialtyService

EvaluateCareFacade (application/facades/evaluate_care_facade.py)

Methods:

- search_clinical_trials(input: ClinicalTrialsInput) -> ClinicalTrialsOutput
- get_provider_details(npi: str) -> ProviderDetailOutput

Delegates to: ClinicalTrialsService, ProviderDetailService

Cross-calls: FindCareFacade.get_provider_location()

7. Tool Registry (ToolRouter)

Class: ToolRouter
| File: application/tool_router.py
| Pattern: Pydantic-validated allowlist dispatch

Tool Name	Pydantic Model	Handler
find_providers	ProviderSearchInput	FindCareFacade.search_providers
find_specialty_codes	SpecialtyInput	FindCareFacade.identify_specialty
search_clinical_trials	ClinicalTrialsInput	EvaluateCareFacade.search_clinical_trials
lookup_provider_external	ProviderDetailInput	EvaluateCareFacade.get_provider_details
get_skip_snow_context	None	AboutService.get_skip_snow_context
get_chathealthy_context	None	AboutService.get_chathealthy_context
request_consent	ConsentRequestInput	ConsentService.request_consent
record_verbatim_consent	VerbatimConsentInput	ConsentService.record_verbatim_consent
record_summary_consent	SummaryConsentInput	ConsentService.record_summary_consent
record_user_details	LeadInput	LeadService.record_user_details
record_unknown_question	UnknownInput	UnknownQuestionService.record
get_provider_location	LocationInput	FindCareFacade.get_provider_location
commit_significant_activity	ActivityInput	LeadService.commit_activity

Enforcement Rules

- Reject unknown tools (F-05 fix -- eliminates globals().get)
- Validate all inputs via Pydantic models
- Validate outputs before returning to LLM
- Server-side enforcement (F-17 fix)

8. Repository File Tree

Code/ConversationalLUX/FindCareChat//

app.py: HuggingFace entrypoint only (~30 lines)

adapter//

host_chat_adapter.py: Translates HTTP <-> ChatOrchestrator

application//

chat_orchestrator.py: LLM loop, system prompt, safety gating

tool_router.py: Pydantic-validated tool dispatch (allowlist enforced)

session_manager.py: Chat session + UX state + timer

facades//

find_care_facade.py: Entry point for all FindCare capabilities

evaluate_care_facade.py: Entry point for all EvaluateCareQuality capabilities

tool_models//

__init__.py:

provider_search_models.py: Pydantic input/output for provider search

clinical_trials_models.py: Pydantic input/output for clinical trials

consent_models.py: Pydantic models for consent workflow (ConsentRequestInput, ConsentState, etc.)

shared_models.py: Common tool schemas (LeadInput, UnknownInput, LocationInput)

domain//

find_care//

provider_search_service.py: Feature 1 -- vector search, query expansion, formatting

specialty_service.py: Feature 2 -- NLP taxonomy mapping

evaluate_care_quality//

clinical_trials_service.py: Feature 3 -- ClinicalTrials.gov integration

provider_detail_service.py: Feature 8 -- NPI lookup, external data

shared//

safety//

safety_service.py: Feature 5 -- IP locking, session safety

emergency_detector.py: F-09 -- config-driven emergency detection

consent//

__init__.py:

consent_service.py: Feature 7 -- two-tier HIPAA consent

de_identify.py: Shared PII scrubbing utility (used by ConsentService + UnknownQuestionService)

lead_capture//

lead_service.py: Feature 6 -- contact record creation

content//

about_service.py: Feature 4 -- about intents

unknowns//

unknown_question_service.py: Feature 12 -- unanswerable question logging

url//

url_guardian.py: Feature 9 -- dual-mode URL validation (IMPLEMENTED)

```
infrastructure//
  mongo//
    provider_repository.py: Read-only provider queries
  external_apis//
    __init__.py:
    npi_client.py: NPI registry lookups
    clinical_trials_client.py: ClinicalTrials.gov API
    google_routes_client.py: Travel time calculations (used by ClinicalTrialsService)
  embeddings//
    embedding_client.py: Vector search client
  config//
    config_loader.py: Loads brain artifacts + environment variables
frontend//
  src//
    components//
      ChatWindow.tsx
      MessageBubble.tsx
      EmergencyBanner.tsx
      TypingIndicator.tsx
    api//
      chat_client.ts
    state//
      chat_store.ts
    rendering//
      markdown_renderer.tsx
    styles//
      chat.css
Code/DataPipelines//
  status: No structural change (GOV-005 preserved)
  security_patch: Add APIKeyMiddleware to status endpoint (F-06)
Code/Shared//
  purpose: Cross-application infrastructure only (NO business logic)
  logging/: Central structured logging
  config/: Shared env loader
  contracts/: Optional shared DTOs (non-domain)
Website//
  status: No change -- remains Cloudflare static (GOV-005)
  note: Remove any leaked architecture docs (F-13)
```

9. Migration Plan (7 Phases)

Strategy: Strangler Fig pattern inside same codebase (incremental extraction, zero downtime)

Phase	Name	Build	Risk	Key Changes
1	Introduce Skeleton Architecture	B-187	LOW	Create folder structure (application/, domain/, infrastructure/, adapter/); Add ChatOrchestrator (empty pass-through); Add ToolRouter with static mapping (still calls existing main.py functions) (+1 more)
2	Extract FindCare -- Provider Search	B-190	LOW	Move find_providers() -> ProviderSearchService.search(); Introduce FindCareFacade.search_providers(); Update ToolRouter to call facade instead of main.py
3	Extract FindCare -- Specialty Service	B-191	LOW-MEDIUM	Move find_specialty_codes() to SpecialtyService; Wire through FindCareFacade; Remove corresponding logic from main.py
4	Extract EvaluateCareQuality Domain	B-193	MEDIUM	Move ClinicalTrialsService; Move ProviderDetailsService; Introduce EvaluateCareFacade (+1 more)
5	Extract Shared Services	B-194	MEDIUM	SafetyService + EmergencyDetector; ConsentService; LeadService (+3 more)
6	Introduce Pydantic Tool Models	B-195	MEDIUM	Define all tool input/output schemas; Wrap ToolRouter validation layer; Gradually enforce typing per tool
7	Delete main.py Monolith	B-196	LOW (if prior phases complete)	Remove remaining functions from main.py; Ensure all calls go through orchestrator + services; app.py HostChatAdapter is the sole entry point (+1 more)

Safety Mechanisms

- Each phase preserves working state
- Feature-by-feature migration (no big bang)
- Fallback: old functions remain callable until replaced
- Full regression test after Phase 7 (B-197)

10. Testing Strategy

Test Structure

tests/application/

test_chat_orchestrator.py

tests/domain/find_care/

test_provider_search_service.py

test_specialty_service.py

tests/domain/evaluate_care_quality/

test_clinical_trials_service.py

test_provider_detail_service.py

tests/domain/shared/

test_safety_service.py

test_emergency_detector.py

test_consent_service.py

test_lead_service.py

test_unknown_question_service.py

tests/infrastructure/

test_provider_repository.py

test_external_clients.py

Approach

unit_tests: All domain services with mocked dependencies

integration_tests:

- ToolRouter + Facades
- ChatOrchestrator with fake LLM responses

e2e_tests: Real systems (MongoDB, Anthropic) -- no mocks per Boss directive

regression:

- Capture baseline outputs from main.py before refactor
- Replay test prompts and compare outputs

existing_tests: Retained initially, then migrated per service

11. Pydantic Strategy

Scope: All tool inputs and outputs validated via Pydantic models

Location: application/tool_models/

Models

- ProviderSearchInput / ProviderSearchOutput
- SpecialtyInput / SpecialtyOutput
- ClinicalTrialsInput / ClinicalTrialsOutput
- ProviderDetailInput / ProviderDetailOutput
- ConsentRequestInput / ConsentState
- VerbatimConsentInput / VerbatimConsentOutput
- SummaryConsentInput / SummaryConsentOutput
- ConsentAuditEntry
- LeadInput / LeadOutput
- UnknownInput / UnknownOutput
- LocationInput

Enforcement

- ToolRouter validates input before calling handler
- ToolRouter validates output before returning to LLM
- Schema mismatches raise ValidationError (not silently ignored)

12. Perplexity Audit Response (All 18 Findings)

Audit Source: Perplexity external audit v1 (2026-03-28)

Total Findings: 18

Resolved: 8 | Deferred: 6 | Acknowledged: 4 | Strengths: 1

ID	Title	Status	Response
F-01	Three-application separation is clean and well-reasoned	STRENGTH	Three-application boundary rules preserved. GOV-005 enforced throughout refactor.
F-02	Production-grade pipeline engineering	STRENGTH	DataPipelines remains independent. No structural changes. GOV-005 alignment confirmed.
F-03	Safety system is well-designed for a health application	STRENGTH	Safety system preserved in SafetyService + EmergencyDetector. FDA legal review recommended (before Alpha, not code).
F-04	Pipeline framework is correctly designed for AI-operated company	STRENGTH	Pipeline framework unchanged. Correct for Claude-as-developer model.
F-05	globals().get(name) tool dispatch is a security surface	RESOLVED	ToolRouter allowlist replaces globals() dispatch. Phase 1 (B-187).
F-06	Durable Functions status URL exposes orchestration state	DEFERRED	Add APIKeyMiddleware to DataPipelines status endpoint. Pipeline concern, not chat refactor. Deferred to next sprint.
F-07	main.py monolith is a velocity blocker	RESOLVED	main.py eliminated in Phase 7 (B-196). Replaced by modular domain architecture.
F-08	Three-state coverage is thin but rationale is sound	DEFERRED	Pipeline data run, not code change. Deferred to next sprint.
F-09	Emergency keyword list has coverage gaps	RESOLVED	EmergencyDetector with config-driven keywords from brain/machine_artifacts/emergency_keywords.json. Phase 5 + B-189.
F-10	Vector search hard cap of 10 results is too restrictive	RESOLVED	Configurable limit in ProviderSearchService (default 25). B-192.
F-11	FIPS hack should be resolved in next pipeline run	DEFERRED	Pipeline re-run required. Deferred to next sprint.
F-12	DeepSeek disclosure in privacy policy is incomplete	DEFERRED	Legal/copy concern. Deferred to next sprint.
F-13	Architecture documentation is publicly accessible	DEFERRED	Boss decision required on access gate before Alpha.
F-14	Documentation inconsistencies	DEFERRED	Copyright year, broken links, nav button. Low priority, deferred.

F-15	Brain governance system is an effective SDLC	STRENGTH	Brain governance preserved and strengthened. GPT Reader service extends GPT's independent audit capability.
F-16	Brain lock has no deadlock recovery	ACKNOWLEDGED	BrainLockManager with TTL design documented. Implementation deferred to v0.1.4 (BL-003).
F-17	Server-side scope enforcement must be independent of Claude	RESOLVED	ToolRouter + Pydantic validation enforces server-side. No tool can be called without being in the allowlist.
F-18	Audit should cover Claude's actions too	ACKNOWLEDGED	Central structured logging planned in Code/Shared/logging/. Claude audit trail deferred to v0.1.4 (BL-004).

13. Sprint Schedule (Tue-Sun, B-187 through B-201)

Sprint Start: 2026-03-31

Sprint Goal: Refactor architecture (ARCH-001), maps, current functionality enhancements

Day	Theme	Build	Task	Risk	Deploy
2026-03-31 (Tuesday)	Skeleton + Security Fix + Maps	B-187	ARCH-001 Phase 1: Skeleton Architecture + ToolRouter	LOW	dev
2026-03-31 (Tuesday)	Skeleton + Security Fix + Maps	B-188	Maps: Provider search results on map	LOW-MEDIUM	dev
2026-03-31 (Tuesday)	Skeleton + Security Fix + Maps	B-189	F-09: Expand emergency keyword list	LOW	dev
2026-04-01 (Wednesday)	Extract FindCare Domain	B-190	ARCH-001 Phase 2: Extract ProviderSearchService	LOW	dev
2026-04-01 (Wednesday)	Extract FindCare Domain	B-191	ARCH-001 Phase 3: Extract SpecialtyService	LOW-MEDIUM	dev
2026-04-01 (Wednesday)	Extract FindCare Domain	B-192	F-10: Raise provider search result cap to 25	LOW	dev
2026-04-02 (Thursday)	Extract EvaluateCareQuality + Shared	B-193	ARCH-001 Phase 4: Extract ClinicalTrialsService + ProviderDetailService	MEDIUM	dev
2026-04-02 (Thursday)	Extract EvaluateCareQuality + Shared	B-194	ARCH-001 Phase 5: Extract Shared Services	MEDIUM	dev
2026-04-03 (Friday)	Pydantic + Cleanup + QA	B-195	ARCH-001 Phase 6: Pydantic tool models	MEDIUM	dev
2026-04-03 (Friday)	Pydantic + Cleanup + QA	B-196	ARCH-001 Phase 7: Delete main.py	LOW	dev
2026-04-03 (Friday)	Pydantic + Cleanup + QA	B-197	Full regression test + SIT	LOW	dev
2026-04-04 (Saturday)	QA Promotion + Stabilization	B-198	Promote to QA	LOW	qa
2026-04-04 (Saturday)	QA Promotion + Stabilization	B-199	Boss UAT on QA	N/A	qa
2026-04-05 (Sunday)	Bug fixes + Production (if needed)	B-200	Fix any UAT bugs	VARIES	qa
2026-04-05 (Sunday)	Bug fixes + Production (if needed)	B-201	Promote to Production	LOW (if QA passed)	prod

Enhancements Included

- F-05: ToolRouter replaces globals() dispatch (Tuesday (Phase 1))
- F-09: Emergency keyword expansion (Tuesday)
- F-10: Provider search cap raised to 25 (Wednesday)
- MAPS: Interactive map for provider search results (Tuesday)

Deferred to Next Sprint

- F-06: Durable Functions status URL auth

- F-08: All-state coverage (pipeline run)
- F-11: FIPS hack resolution (pipeline re-run)
- F-12: DeepSeek privacy disclosure (legal/copy)
- F-13: Lock architecture docs (Boss decision)
- R19: Azure Function timeout UAT
- R35: Paused cluster UAT
- BL-001 through BL-010: Perplexity backlog (v0.1.4)

14. Architectural Decisions

Facade-oriented domains over flat service layer

Two business domains (FindCare, EvaluateCareQuality) each get a facade. Facades are the ONLY entry point from ToolRouter to domain services. Prevents cross-domain coupling.

Pydantic validation at ToolRouter boundary

Claude sends JSON tool arguments. Pydantic validates at the ToolRouter before reaching domain services. Catches schema mismatches early.

System prompt stays in ChatOrchestrator (not PromptBuilder)

Per Boss decision. Prompt references tool names only (not class names). ToolRouter is source of truth for available tools.

Provider Detail belongs to EvaluateCareQuality (not FindCare)

Provider detail is a quality evaluation concern (NPI lookup, board certifications, ratings). Finding a provider and evaluating their quality are distinct business actions.

Frontend remains thin -- no domain duplication

All domain logic stays server-side. Frontend is presentation-only: ChatWindow, MessageBubble, EmergencyBanner. No backend domain structure mirrored.

Strangler Fig migration (not big bang)

7 phases, each deployable. Old functions remain callable until replaced. Zero downtime throughout migration.

No dependency injection for MongoDB/notification clients

Boss rule: e2e tests use real systems, no mocks. DI for testability is unnecessary when you do not mock.

15. Governance Alignment

Policy	Status
GOV-001	No provider prioritization logic in services
GOV-002	Single-tenant config enforced in Config
GOV-004	LLM suggests, ToolRouter enforces execution
GOV-005	Strict three-application separation preserved

16. Sign-Off

CEO / Final Authority

Status: APPROVED | Date: 2026-03-30

Notes: Design and sprint plan approved. Consent elevated to bounded context per Boss directive. Appendix A (AI model usage) approved. Proceed to build.

Engineer: Claude Opus 4.6

Status: APPROVED | Date: 2026-03-30

Notes: Design is fully executable. 7-phase migration is incremental and safe. All 18 Perplexity findings addressed.

Architect: GPT-5.3

Status: APPROVED | Date: 2026-03-30

Notes: Architecture is sound. Facade pattern correctly separates domains. Pydantic enforcement at tool boundary is the right pattern.

Appendix A: AI Model Usage by Prompt

Every AI prompt in the system with the model used, its purpose, and cost profile.

Runtime Prompts (Per User Session)

Prompt	Model	Purpose	Invocation
Chat conversation (user-facing)	Claude Sonnet 4.6 (Anthropic)	Main conversational AI -- tool selection, response generation, intent routing	Every user message
Safety classifier (emergency detection)	GPT-4.1-mini (OpenAI)	Semantic emergency classifier -- second trigger in dual-trigger OR logic	Every user message (parallel with keyword check)
Vector embedding (provider search)	text-embedding-3-large (OpenAI)	Embed user query for vector similarity search against provider descriptions	Every provider search tool call
Vector embedding (specialty search)	text-embedding-3-large (OpenAI)	Embed specialty query for NUCC code matching	Every specialty identification tool call
De-identification (consent flow)	Claude Haiku 4.5 (Anthropic)	Summarize and strip PII from conversation transcript	Only when user declines verbatim but accepts summary consent
Conversation summarization (consent flow)	Claude Haiku 4.5 (Anthropic)	Generate summary of conversation before de-identification	Only when Tier 2 consent granted

Pipeline Prompts (Batch)

Prompt	Model	Purpose	Invocation
Provider description embedding (batch)	text-embedding-3-large (OpenAI)	Generate 3072-dim embeddings for provider vector search index	Pipeline run only (batch, 8.9M records)
County enrichment (geocoding)	N/A (Google Maps API, not AI)	Reverse geocode lat/lng to county FIPS	Pipeline run only

Brain Prompts (Claude + GPT Collaboration)

Prompt	Model	Purpose	Invocation
GPT architecture review / SIT / design collaboration	GPT-5.3-chat-latest (OpenAI)	Enterprise architect role -- design proposals, SIT execution, code audit	On-demand by Claude via brain runner
GPT assurance loop (assignment review)	GPT-5.3-chat-latest (OpenAI)	Review Claude's work against assignments, produce assurance output	Per brain assignment

Model Selection Rationale

Model	Rationale
Claude Sonnet 4.6	Best tool-use performance for conversational AI. Native streaming. Anthropic SDK integration.
GPT-5.3-chat-latest	Reasoning model for architecture and QA. Replaced GPT-4o and o3 after SIT testing showed GPT-4o cannot self-correct in iterative loops and o3 was cost-inefficient (3.3M tokens for 32 iterations). See FIND-SIT-001.
GPT-4.1-mini	Cheapest OpenAI model sufficient for binary safety classification. Called on every message -- cost sensitivity is critical.
Claude Haiku 4.5	Fast, cheap, sufficient for summarization and de-identification. Only invoked during consent flow (rare path).
text-embedding-3-large	3072-dim embeddings for high-quality vector search. OpenAI's best embedding model. Used for both runtime queries and batch pipeline.

Appendix B: PDF-to-JSON Traceability Matrix

Compliance check: every section in the PDF maps to a JSON key. No orphan sections, no missing data.

Section Traceability

PDF Section	JSON Key	Status
1. Executive Summary	executive_summary	TRACED
2. Top-Level Architecture	top_level_architecture	TRACED
3. Business Components	business_components	TRACED
3.1 FindCareFacade	business_components.find_care	TRACED
3.2 EvaluateCareFacade	business_components.evaluate_care_quality	TRACED
4. Shared Infrastructure Services	shared_infrastructure	TRACED
4.1 Consent Framework	consent_flow	TRACED
5. HuggingFace Surface Contract	huggingface_surface	TRACED
6. Facade Design	facade_design	TRACED
7. Tool Registry (ToolRouter)	tool_registry	TRACED
8. Repository File Tree	repo_structure	TRACED
9. Migration Plan	migration_plan	TRACED
10. Testing Strategy	testing_strategy	TRACED
11. Pydantic Strategy	pydantic_strategy	TRACED
12. Perplexity Audit Response	perplexity_audit_response	TRACED
13. Sprint Schedule	sprint_schedule	TRACED
14. Architectural Decisions	architectural_decisions	TRACED
15. Governance Alignment	governance_alignment	TRACED
16. Sign-Off	sign_off	TRACED
Appendix A: AI Model Usage	appendix_ai_models	TRACED
Appendix B: Traceability Matrix	appendix_traceability	TRACED

UAT Feature Traceability

#	Feature	Class	Tool	Test
1	Provider Search	ProviderSearchService	find_providers	test_provider_search_service.py
2	Specialty Identification	SpecialtyService	find_specialty_codes	test_specialty_service.py
3	Clinical Trials Search	ClinicalTrialsService	search_clinical_trials	test_clinical_trials_service.py
4	About ChatHealthy	AboutService	get_skip_snow_context / get_chathealthy_context	N/A (static content)
5	Safety Filter	SafetyService + EmergencyDetector	N/A (pre-tool gating)	test_safety_service.py + test_emergency_detector.py
6	Lead Capture	LeadService	record_user_details	test_lead_service.py
7	Consent Framework	ConsentService	request_consent / record_verbatim_consent / record_summary_consent	test_consent_service.py
8	Provider Detail	ProviderDetailService	lookup_provider_external	test_provider_detail_service.py
9	URL Guardian	URLGuardian	N/A (post-response filter)	N/A (existing tests)
10	Chat UX	SessionManager	N/A (application layer)	test_chat_orchestrator.py
11	Blob Storage	BlobStorageClient	N/A (infrastructure)	N/A (pipeline concern)
12	Unanswerable Questions	UnknownQuestionService	record_unknown_question	test_unknown_question_service.py
13	Markdown Rendering	N/A	N/A (frontend only)	N/A (frontend)

Compliance Result

ALL TRACED -- 21 PDF sections map to JSON keys. 13 UAT features traced to classes, tools, and tests. 18 Perplexity findings traced to JSON responses. Zero orphans.

End of Document